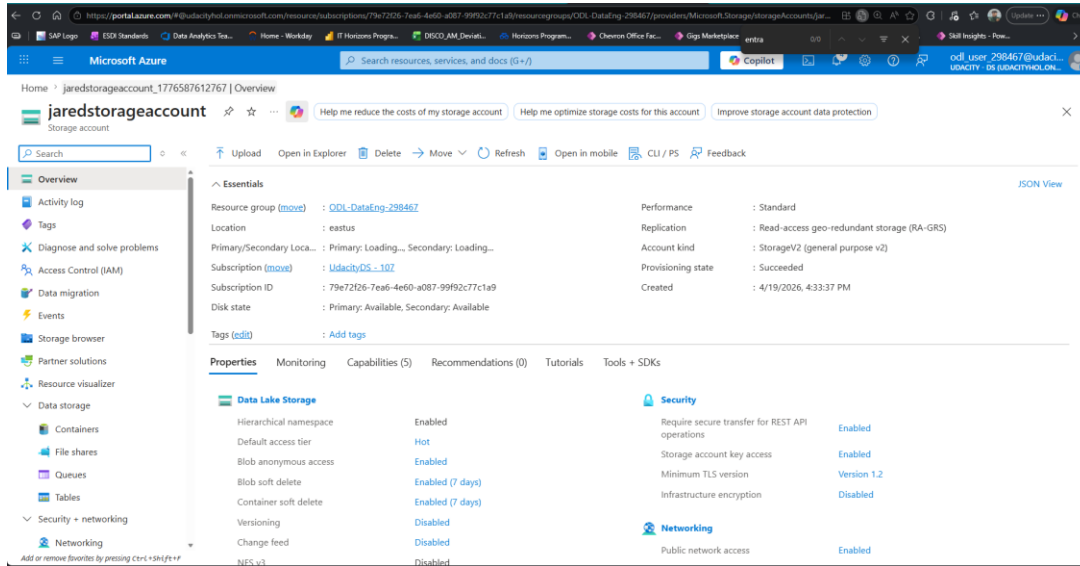
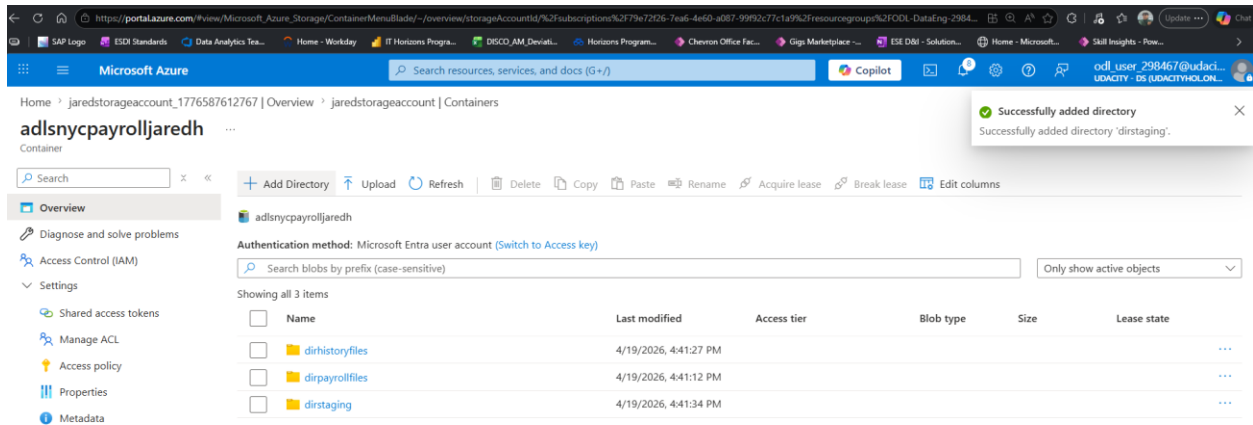


Create Storage Account:



Create an Azure Data Lake Storage Gen2 (storage account) and associated storage container resource named adlsnycpayroll-yourfirstname-lastinitial. Create three directories in this storage container



Home > jaredstorageaccount_1776587612767 | Overview > jaredstorageaccount | Containers

adlsnycpayrolljaredh

Container

Search

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 📄 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

- Shared access tokens
- Manage ACL
- Access policy
- Properties
- Metadata

Authentication method: Microsoft Entra user account (Switch to Access key)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 4 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[...]					...
☐	AgencyMaster.csv	4/19/2026, 4:42:59 PM	Hot (Inferred)	Block blob	4.9 KiB	Available
☐	EmpMaster.csv	4/19/2026, 4:42:59 PM	Hot (Inferred)	Block blob	23.04 KiB	Available
☐	TitleMaster.csv	4/19/2026, 4:43:00 PM	Hot (Inferred)	Block blob	49.04 KiB	Available
☐	nycpayroll_2021.csv	4/19/2026, 4:42:59 PM	Hot (Inferred)	Block blob	16.45 KiB	Available

Home > jaredstorageaccount_1776587612767 | Overview > jaredstorageaccount | Containers

adlsnycpayrolljaredh

Container

Search

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 📄 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

- Shared access tokens
- Manage ACL
- Access policy
- Properties
- Metadata

Authentication method: Microsoft Entra user account (Switch to Access key)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[...]					...
☐	nycpayroll_2020.csv	4/19/2026, 4:44:13 PM	Hot (Inferred)	Block blob	17.07 KiB	Available

Create Azure Data Factory

Home > Microsoft.DataFactory-20260419164641 | Overview > ODL-DataEng-298467

jaredhdatafactory

Data factory (V2)

How do I troubleshoot pipeline failures in this Data factory (V2)? | List triggers associated with this Data factory (V2) | Show me integration runtime metrics for this Data factory (V2)

Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Monitoring

Automation

Help

Essentials

Resource group (broad) : ODL-DataEng-298467

Status : Succeeded

Location : East US

Subscription (broad) : UdaCity-DS-107

Subscription ID : 79e72d6-7ea6-4e60-a087-9992c77c1a9

Type : Data factory (V2)

Getting started : Quick start

Azure Data Factory Studio

Launch studio

Quick Starts

Tutorials

Template Gallery

Training Modules

Create SQL DB

The screenshot shows the Microsoft Azure portal interface for a newly created SQL database. The browser address bar displays the URL: https://portal.azure.com/#@udacityhol.onmicrosoft.com/resource/subscriptions/79e72f26-7ea6-4e60-a087-99f92c77c1a9/resourceGroups/ODL-DataEng-298467/providers/Microsoft.Sql/servers/jaredatabaseserver/db_nycpayroll. The page title is "db_nycpayroll (jaredatabaseserver/db_nycpayroll) SQL database".

The left sidebar contains a navigation menu with the following items: Overview (selected), Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, Intelligent performance, Monitoring, Automation, and Help.

The main content area is titled "Essentials" and displays the following details:

- Resource group (move): [ODL-DataEng-298467](#)
- Status: Online
- Location: East US
- Subscription (move): [UdacityDS - 107](#)
- Subscription ID: 79e72f26-7ea6-4e60-a087-99f92c77c1a9
- Tags (edit): [Add tags](#)
- Server name: [jaredatabaseserver.database.windows.net](#)
- Elastic pool: [No elastic pool](#)
- Connection strings: [Show database connection strings](#)
- Pricing tier: Basic
- Earliest restore point: No restore point available

Below the details, there are tabs for "Getting started", "Monitoring", "Properties", "Features", "Notifications (0)", "Integrations", and "Tutorials". The "Getting started" tab is active, showing a section titled "Start working with your database" with the instruction: "Connect to your database and start working with data with a few simple steps. [Learn more](#)".

Four action cards are displayed:

- Configure access**: Configure network access to your SQL server. [Learn more](#)
- Connect to application**: Use connection strings to connect to your SQL database from your applications and favorite tools.
- Start developing**: Work in your database by using tools to add, modify and query data. [Compare tools](#)
- Mirror database in Fabric**: Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

Create Data Source:

The screenshot shows the Microsoft Synapse Studio Query Editor interface. The top bar includes a file explorer with "SQL script 1" and "adlsnycpayrolljaredh", a warning message "Other users in your workspace may have access to modify this item. Do not use this who may have access to the workspace.", and buttons for "Run", "Undo", "Publish", and "Query plan". The "Connect to" dropdown is set to "Built-in" and the "Use database" dropdown is set to "udacity".

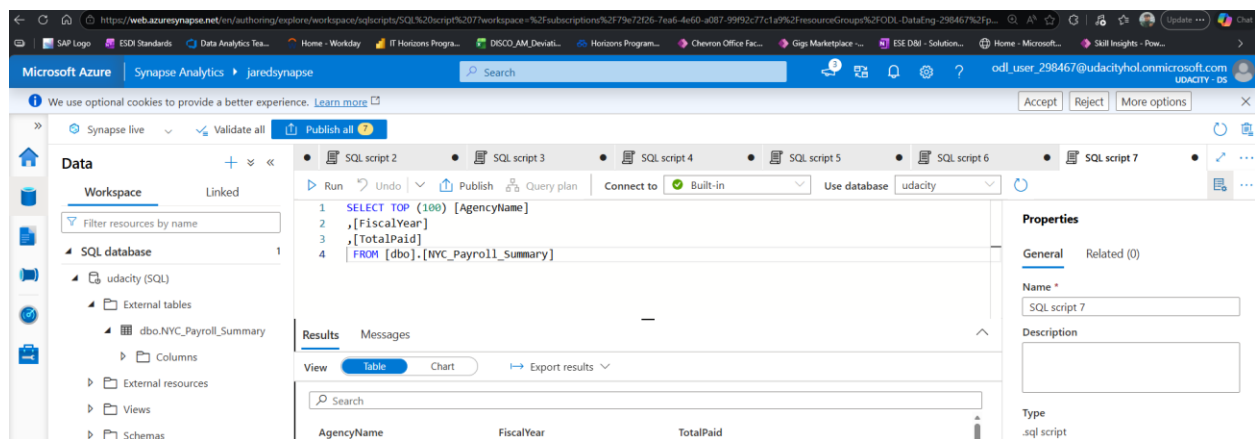
The SQL script in the editor is as follows:

```
1
2 -- 2) Create External Data Source that points directly to /dirstaging
3 CREATE EXTERNAL DATA SOURCE MyNYCPayrollDS
4 WITH (
5     LOCATION = 'abfss://adlsnycpayrolljaredh@jaredstorageaccount.dfs.core.windows.net/dirstaging'
6     -- CREDENTIAL = <YourCredentialName> -- add if your Synapse requires it
7 );
8 GO
```

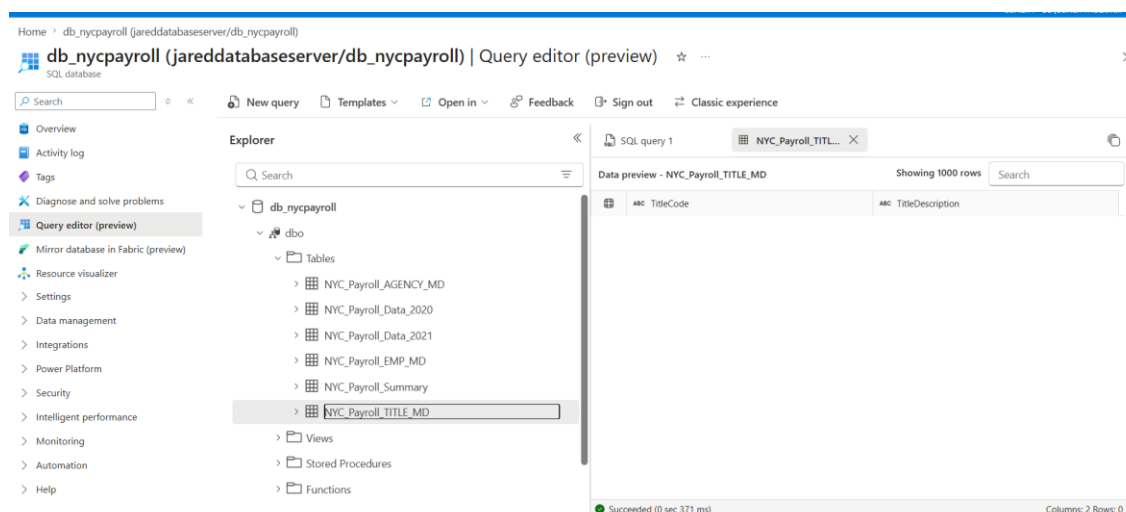
The bottom section of the interface shows the "Results" tab, which is currently empty with the message "No results to show". The "Messages" tab shows a status message: "00:00:09 Query executed successfully."

Create External Table

```
2 -- Recreate it with column order matching the file
3 CREATE EXTERNAL TABLE dbo.NYC_Payroll_Summary
4 (
5     AgencyName NVARCHAR(4000),
6     FiscalYear BIGINT,
7     TotalPaid FLOAT
8 )
9 WITH
10 (
11     LOCATION = '/dirstaging/',
12     DATA_SOURCE = [abfss://adlsnycpayrolljaredh@jaredstorageaccount.dfs.core.windows.net],
13     FILE_FORMAT = [SynapseDelimitedTextFormat]
14 );
15 GO
```

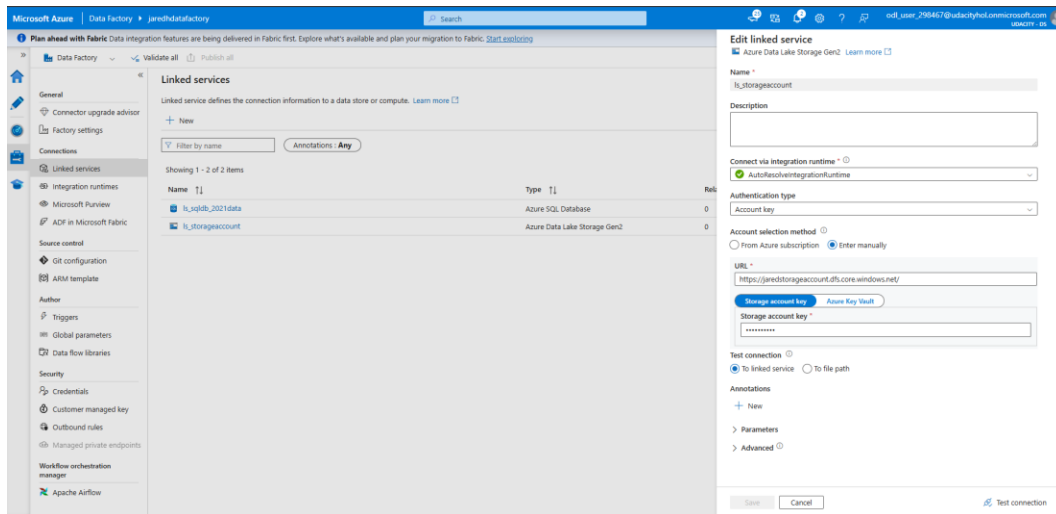


Create Tables in DB



Step 2: Create Linked Services

1. Create a Linked Service for Azure Data Lake



2. Create a Linked Service to SQL Database that has the current (2021) data

New linked service
 Azure SQL Database [Learn more](#)

Version
☒ 2.0 (Recommended) ☐ 1.0
[Import from connection string](#)

Account selection method
☒ From Azure subscription ☐ Enter manually

Azure subscription
 UdacityDS - 107 (79e72f26-7ea6-4e60-a087-99f92c77c1a9)

Server name *
 jaredatabaseserver

Database name *
 db_nycpayroll

Authentication type *
 SQL authentication

User name *
 adminjared

Password *
 Password [Azure Key Vault](#)

Always encrypted ☐

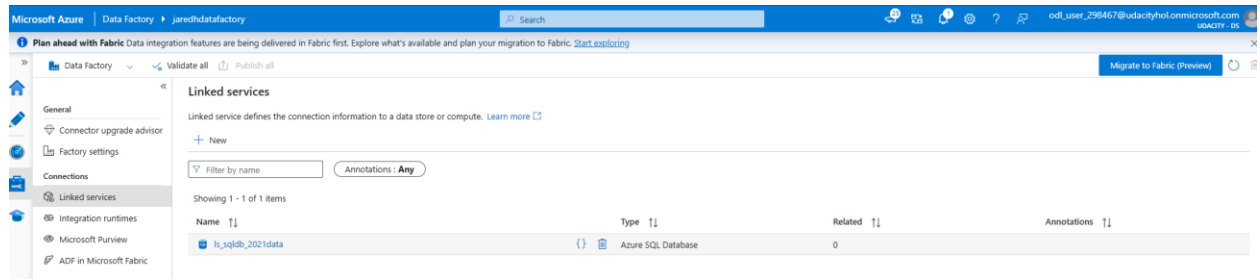
Encrypt
 Mandatory

Trust server certificate ☐

Test name in certificate

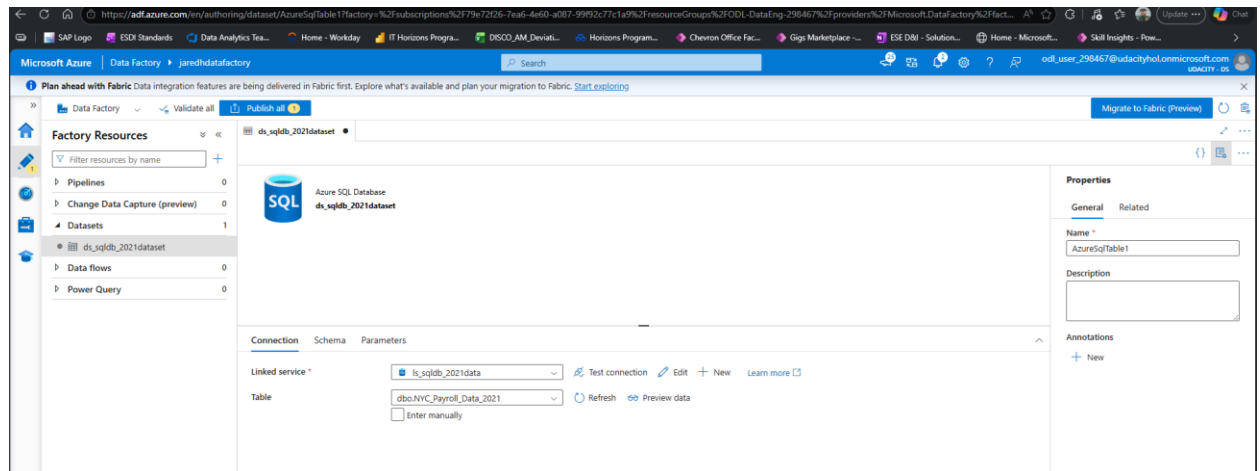
Test connection ☒ Connection successful

[Create](#) [Back](#) [Test connection](#) [Cancel](#)

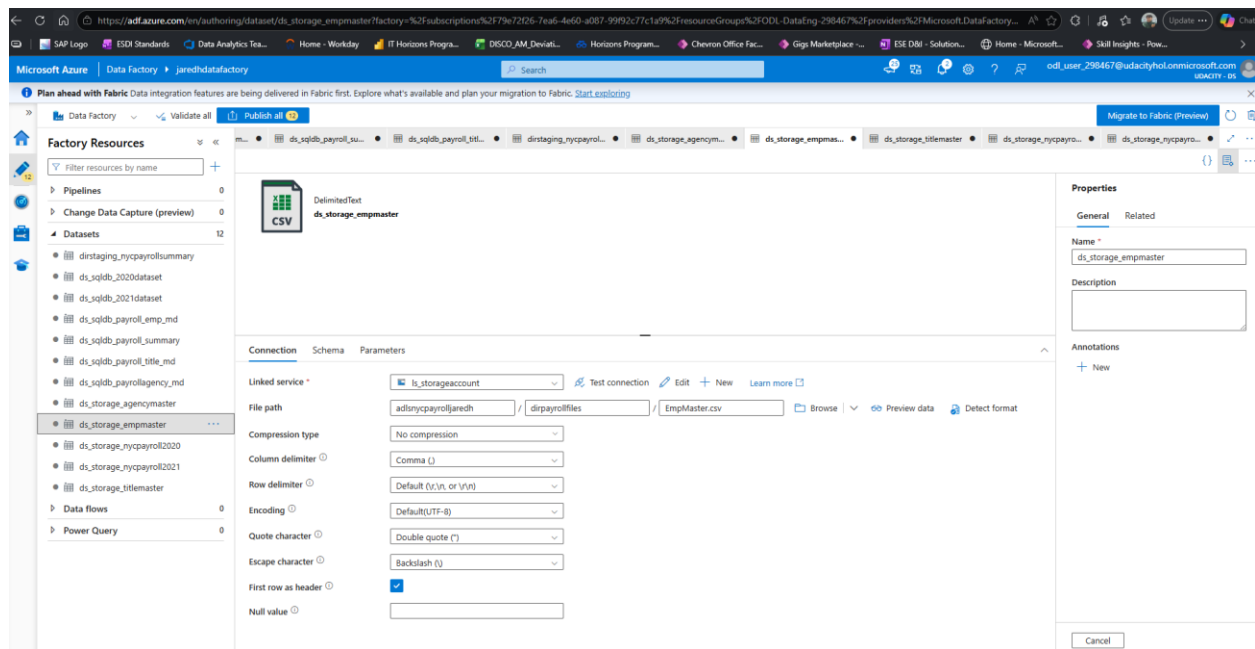


Step 3: Create Datasets in Azure Data Factory

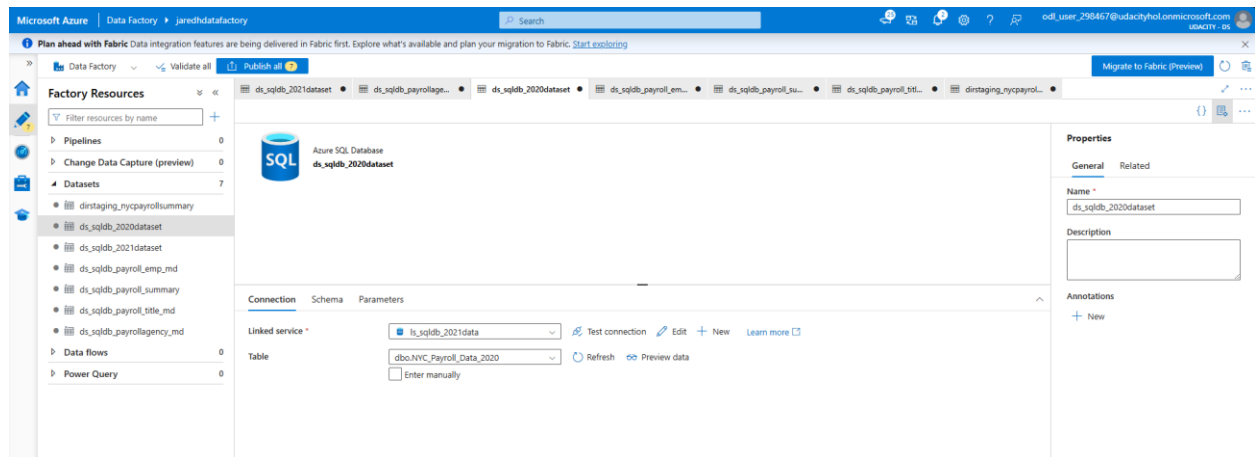
1.Create the datasets for the 2021 Payroll file on Azure Data Lake Gen2



Repeat the same process to create datasets for the rest of the data files in the Data Lake

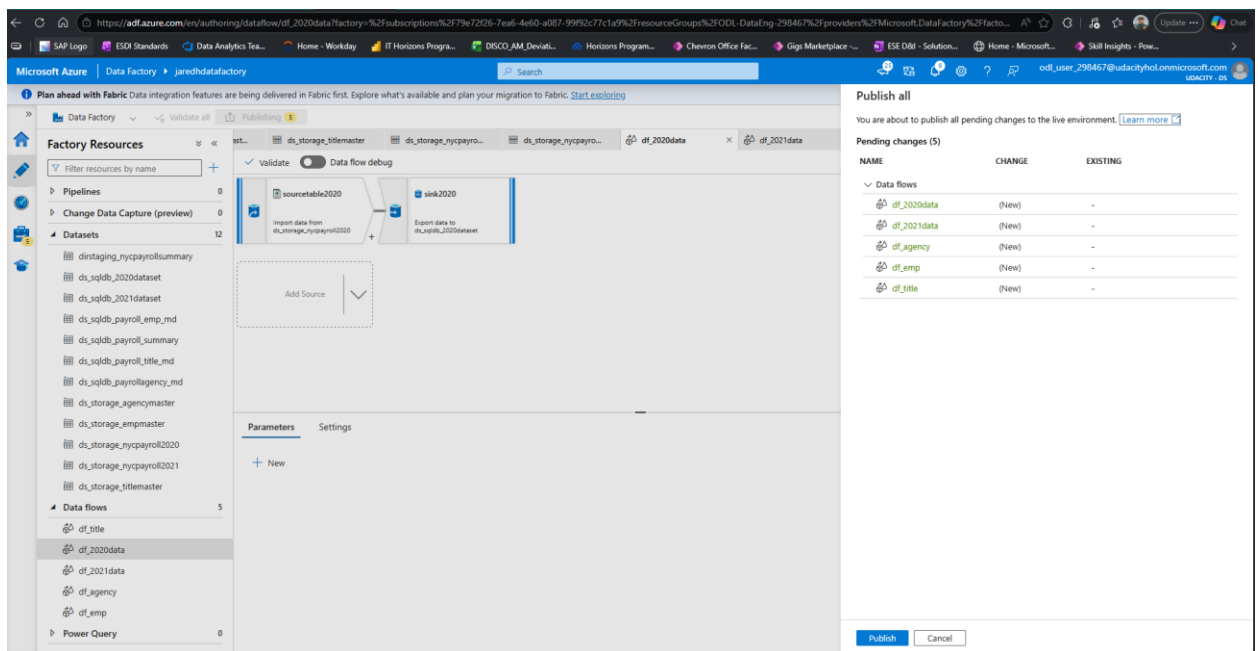
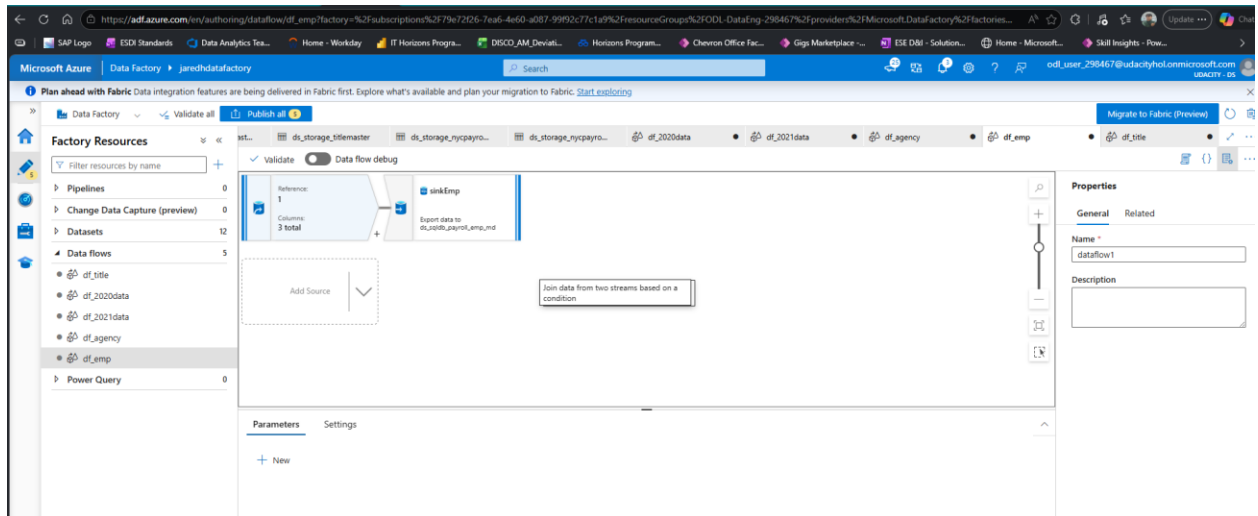


Create the dataset for all the data tables in SQL DB



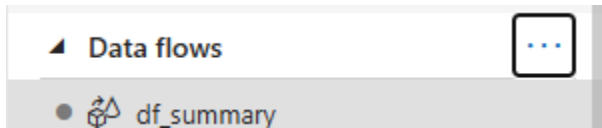


Step 4: Create Data Flows

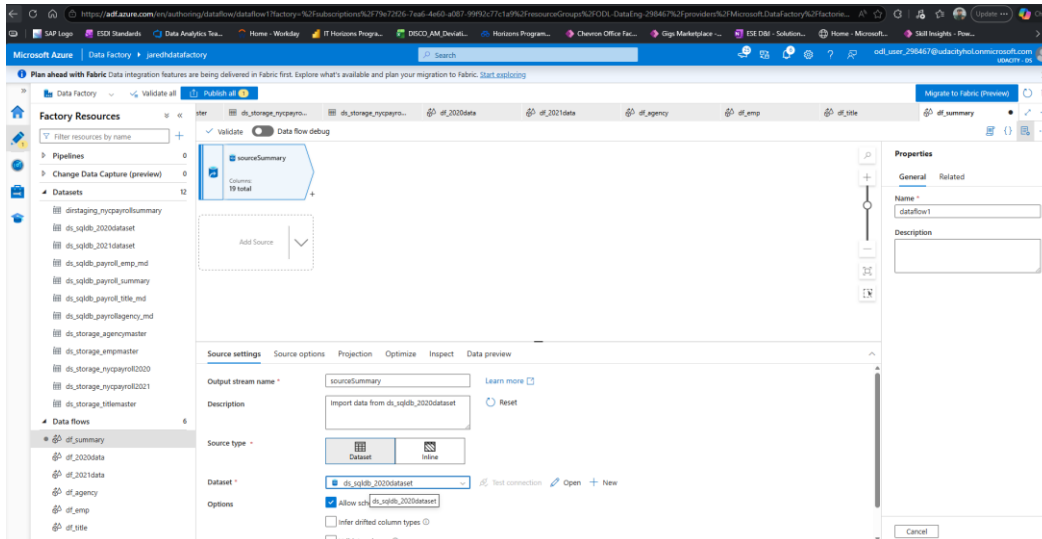


Step 5: Data Aggregation and Parameterization

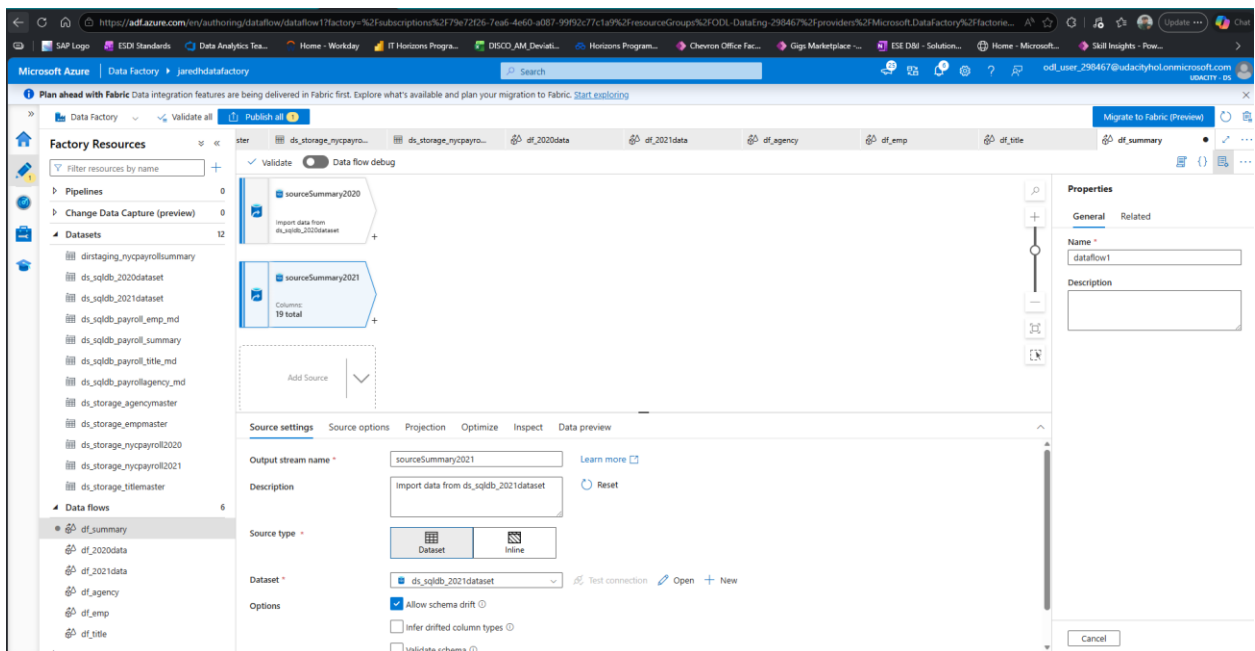
1. Create new data flow and name it Dataflow Summary



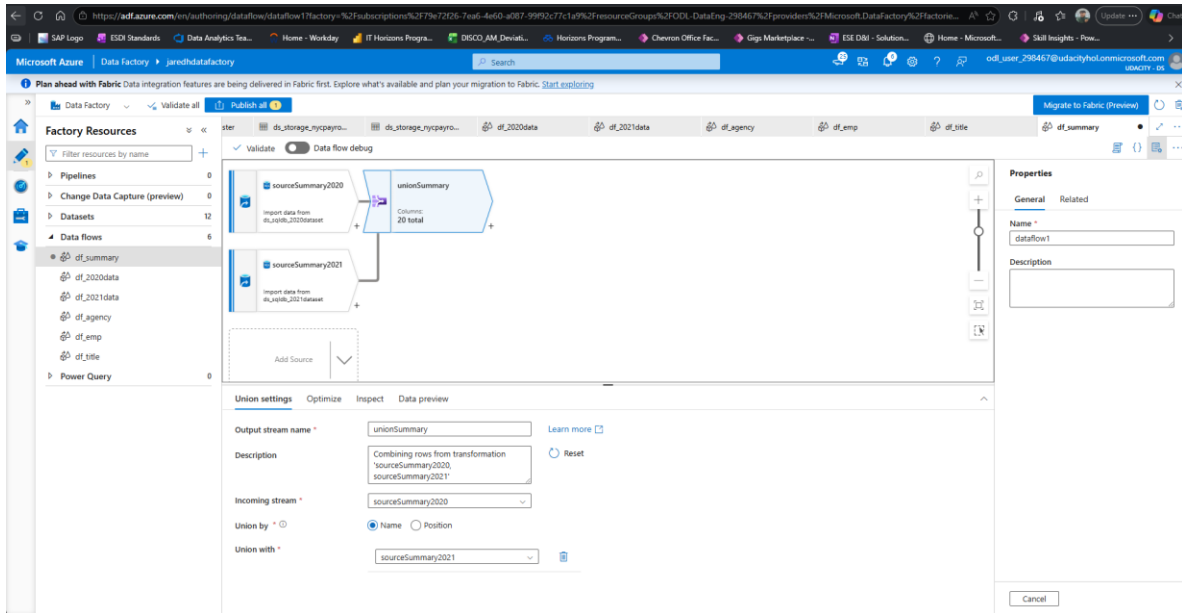
2. Add source as payroll 2020 data from SQL DB



3. Add another source as payroll 2021 data from SQL DB



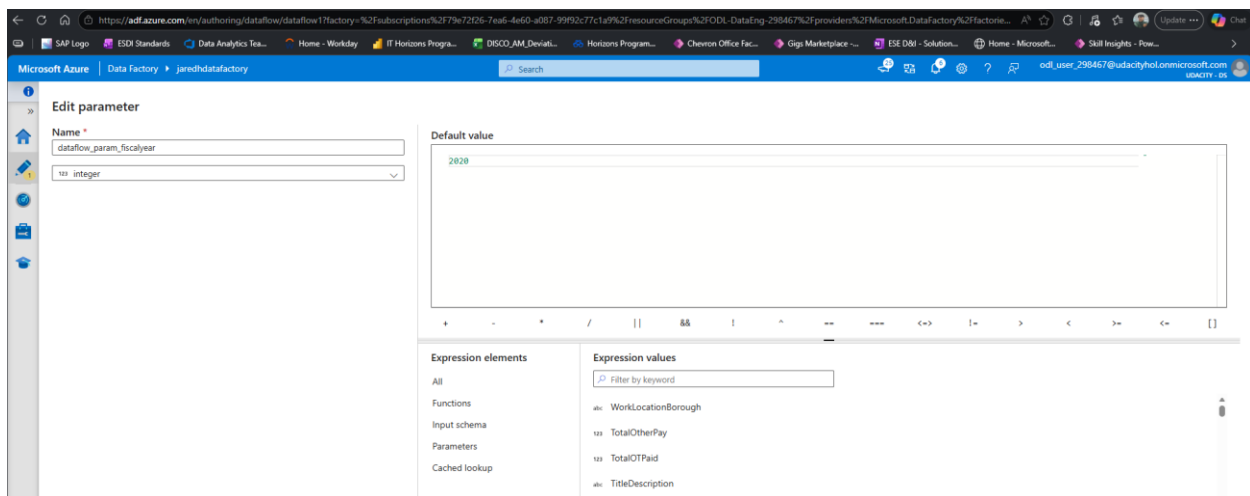
4. Create a new Union activity and select both payroll datasets as the source



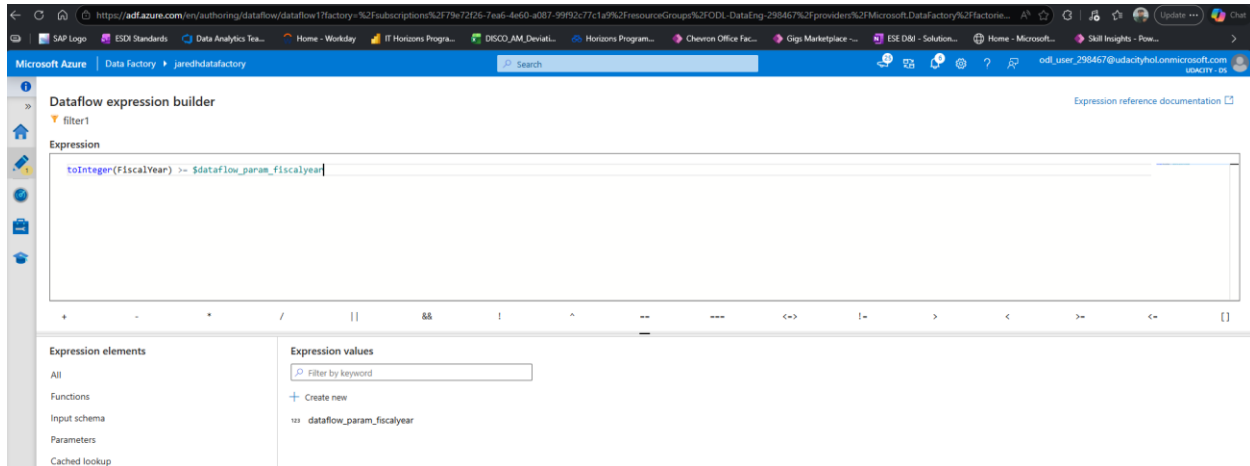
5. Make sure to do any source to target mappings if required. This can be done by adding a Select activity before Union

6. After Union, add a Filter activity, go to Expression builder

a. Create a parameter named- dataflow_param_fiscalyear and give value 2020 or 2021



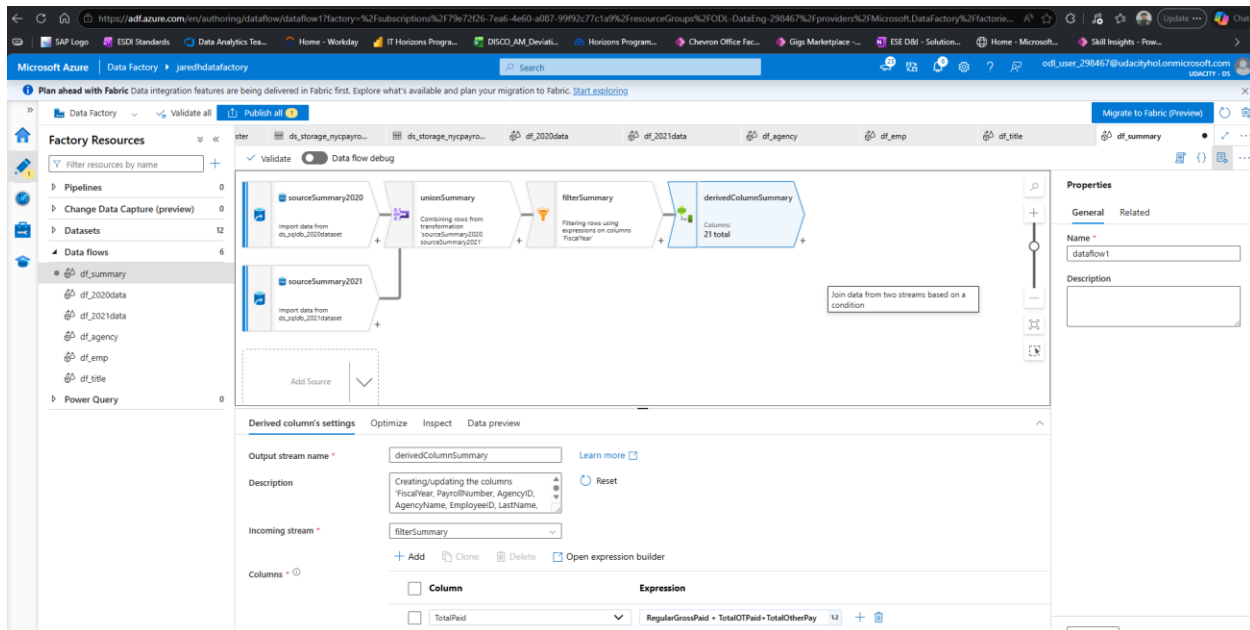
- b. Include expression to be used for filtering: `toInteger(FiscalYear) >= $dataflow_param_fiscalyear`



7. Now, choose Derived Column after filter

- a. Name the column: TotalPaid

- b. Add following expression: `RegularGrossPaid + TotalOTPaid+TotalOtherPay`



8. Add an Aggregate activity to the data flow next to the TotalPaid activity

a. Under Group by, select AgencyName and FiscalYear

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists various resources, including 'Data flows'. The main canvas displays a data flow with several activities: 'sourceSummary2020', 'unionSummary', 'filterSummary', 'derivedColumnSummary', and 'aggregateSummary'. The 'aggregateSummary' activity is selected, and its properties are shown on the right. The 'Aggregate settings' tab is active, showing the 'Output stream name' as 'aggregateSummary', the 'Description' as 'Aggregating data by 'AgencyName, FiscalYear' producing columns 'TotalPaid'', and the 'Incoming stream' as 'derivedColumnSummary'. Under the 'Group by' section, 'AgencyName' and 'FiscalYear' are selected. The 'Properties' pane on the right shows the 'Name' as 'dataflow1' and the 'Description' as 'Join data from two streams based on a condition'.

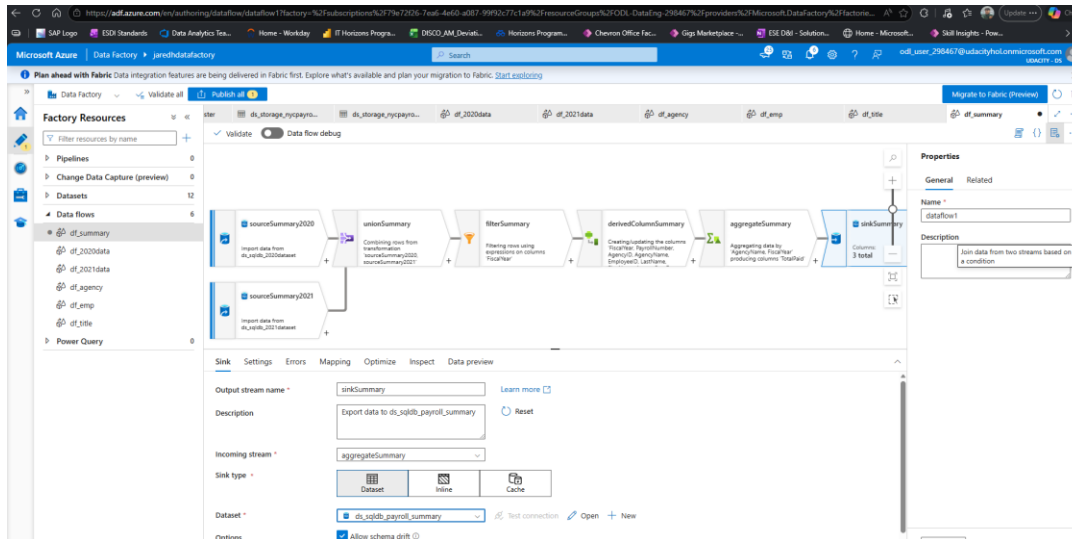
b. Set the expression to sum(TotalPaid)

The screenshot shows the 'Aggregate settings' tab in the Microsoft Azure Data Factory interface. The 'Description' field contains the text 'Aggregating data by 'AgencyName, FiscalYear' producing columns 'TotalPaid''. The 'Incoming stream' is set to 'derivedColumnSummary'. The 'Group by' section shows 'AgencyName' and 'FiscalYear' selected. Below this, there is a table with columns 'Column' and 'Expression'. The 'Column' column has a dropdown menu with 'TotalPaid' selected. The 'Expression' column has a text box containing 'sum(TotalPaid)'.

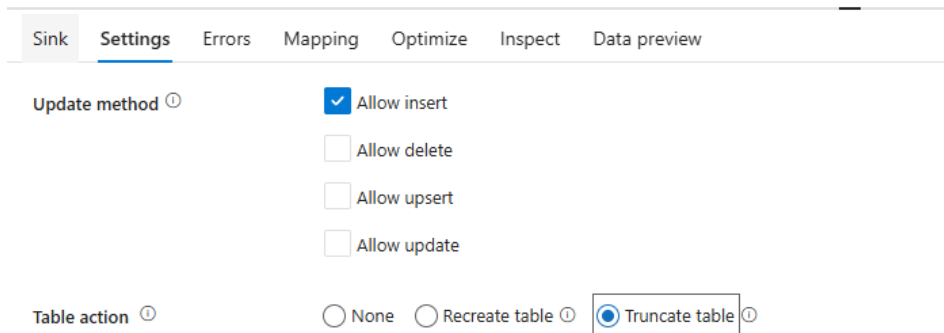
Column	Expression
TotalPaid	sum(TotalPaid)

9. Add a Sink activity after the Aggregate

a. Select the sink as summary table created in SQL db

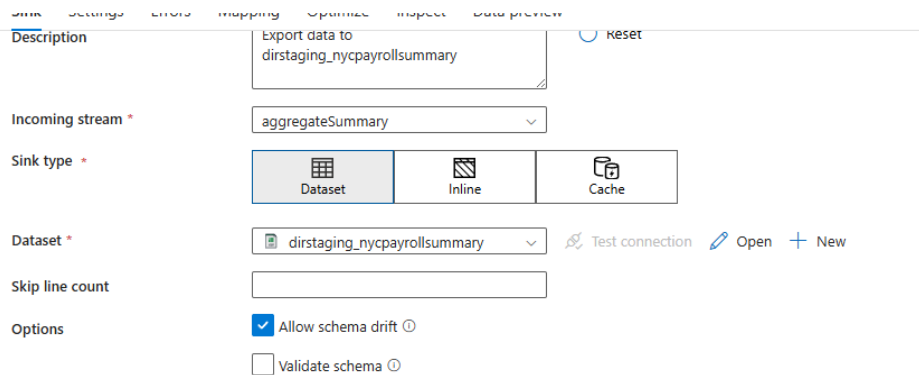


b. In Settings, tick Truncate table



10. Add another Sink activity, this will create two sinks after Aggregate

a. Select the sink as dirstaging in Azure DataLake Gen2 storage



b. In Settings, tick Clear the folder

Sink	Settings	Errors	Mapping	Optimize	Inspect	Data preview
Clear the folder	☒					
File name option *	Default					
Quote All ①	☐					
Headers ①	Enter expression... ANY					
Umask ①	Owner ☐ R ☐ W ☐ X Group ☐ R ☒ W ☐ X Others ☐ R ☒ W ☐ X Octal 022					

Step 6: Pipeline Creation

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines and data flows. The main canvas shows a data flow diagram with activities like 'Data flow' and 'Data flow debug'. The right sidebar shows the 'Properties' panel for the selected pipeline.

Step 7: Trigger and Monitor Pipeline

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

All pipeline runs > pipelineProject - Activity runs

Run > Cancel > Refresh > Update pipeline > List > Gantt

Activity runs

Pipeline run ID: f933eae1-9c9e-4cb1-90c1-3567de272433

All status > Monitor in Azure Metrics > Export to CSV >

Showing 1 - 6 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
df_summary	Succeeded	Data flow	4/19/2026, 10:46:17 PM	1m 5s	AutoResolveIntegrationRuntime (East US)		52ef2456-58cc-4601-a733-e040570702ef	
df_2020data	Succeeded	Data flow	4/19/2026, 10:45:46 PM	30s	AutoResolveIntegrationRuntime (East US)		21b66999-5db1-4c9e-9db1-83161cc826f5	
df_2021data	Succeeded	Data flow	4/19/2026, 10:45:46 PM	29s	AutoResolveIntegrationRuntime (East US)		0a84e5bf-e26c-47e7-af9a-686dca147688	
df_title	Succeeded	Data flow	4/19/2026, 10:40:43 PM	4m 44s	AutoResolveIntegrationRuntime (East US)		596e5914-b88b-4a9c-8dca-b90e5f3d8a41	
df_agency	Succeeded	Data flow	4/19/2026, 10:40:43 PM	5m 3s	AutoResolveIntegrationRuntime (East US)		cc787485-91a8-4b87-ba2d-a26cc0ffa3f	
df_emp	Succeeded	Data flow	4/19/2026, 10:40:43 PM	4m 40s	AutoResolveIntegrationRuntime (East US)		604ab046-cbd8-49ff-8632-9807e95f286a	

Step 8: Verify Pipeline run artifacts

Home > db_nycpayroll (jaredatabaseserver/db_nycpayroll)

db_nycpayroll (jaredatabaseserver/db_nycpayroll) | Query editor (preview)

SQL database

Search resources, services, and docs (G+)

Copilot

Home > db_nycpayroll (jaredatabaseserver/db_nycpayroll)

db_nycpayroll (jaredatabaseserver/db_nycpayroll) | Query editor (preview)

SQL database

Search

New query > Templates > Open in > Feedback > Sign out > Classic experience

Explorer

db_nycpayroll

- dbo
 - Tables
 - NYC_Payroll_AGENCY_MD
 - NYC_Payroll_Data_2020
 - NYC_Payroll_Data_2021
 - NYC_Payroll_EMP_MD
 - NYC_Payroll_Summary
 - NYC_Payroll_TITLE_MD
 - Views
 - Stored Procedures
 - Functions

SQL query 1 > SQL query 2

Run > Save as view

Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your

```
1 SELECT TOP (1000) * FROM [dbo].[NYC_Payroll_Summary]
```

Messages > Results > > >

	FiscalYear	AgencyName	TotalPaid
1	2021	DEPT OF ENVIRONMENT PROTECTION	858802.4
2	2020	OFFICE OF EMERGENCY MANAGEMENT	4348234.9600000003
3	2021	DEPT OF CITYWIDE ADMIN SVCS	1873220.3900000001
4	2021	DEPT OF PARKS & RECREATION	265244.83
5	2020	OFFICE OF MANAGEMENT & BUDGET	2489165.3200000003
6	2021	OFFICE OF THE MAYOR	284742.27999999997

Succeeded (0 sec 441 ms) Columns: 3 Rows: 25

adlsnycpayrolljaredh

SQL script 1

SQL script 2

Other users in your workspace may have access to modify this item. Do not use this item unless you trust all users who may have access to the workspace.

New SQL script

Upload

Download

New folder

Select all

Rename

Manage access

More

adlsnycpayrolljaredh > dirstaging

Name	Last Modified	Content Type	Size
_SUCCESS	4/19/2026, 10:47:18 PM		
part-00000-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:09 PM		32 B
part-00005-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:09 PM		84 B
part-00011-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:09 PM		86 B
part-00023-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:10 PM		77 B
part-00044-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:11 PM		74 B
part-00055-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:12 PM		74 B
part-00060-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:12 PM		86 B
part-00065-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:12 PM		76 B
part-00070-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:12 PM		73 B
part-00077-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:12 PM		78 B
part-00083-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:13 PM		73 B
part-00099-d4f4527d-c5c2-4aa5-a8d2-a9f2aa1636d1-c000.csv	4/19/2026, 10:47:13 PM		75 B

Data in External Table

Microsoft Azure

Synapse Analytics

jaredsynapse

Search

odl_user_298467@udacityhol.onmicrosoft.com

UDACITY · DS

We use optional cookies to provide a better experience. [Learn more](#)

Accept

Reject

More options

Synapse live

Validate all

Publish all

Data

Workspace

Linked

Filter resources by name

SQL database

1

udacity (SQL)

External tables

dbo.NYC_Payroll_Summary

Columns

External resources

Views

Schemas

Security

SQL script 2

SQL script 3

SQL script 4

SQL script 5

SQL script 6

SQL script 7

Run

Undo

Publish

Query plan

Connect to

Built-in

Use database

udacity

1 SELECT TOP (100) [AgencyName]

2 ,[FiscalYear]

3 ,[TotalPaid]

4 FROM [dbo].[NYC_Payroll_Summary]

Results

Messages

View

Table

Chart

Export results

Search

AgencyName	FiscalYear	TotalPaid
DEPT OF ENVIRONMENT PROTE...	2021	2576407.2
NYC HOUSING AUTHORITY	2021	8031325.65
DISTRICT ATTORNEY-MANHATT...	2021	823311.03
BOARD OF ELECTION	2021	816717.27
DEPARTMENT OF CORRECTION	2021	9307627.74
OFFICE OF EMERGENCY MANAG...	2020	13044704.880000005

00:00:04 Query executed successfully.

Properties

General

Related (0)

Name *

SQL script 7

Description

Type

.sql script

Size

90 bytes

Results settings per query

First 5000 rows (default)

All rows